

Technology Stack Template

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Layer	Technology	Version	Justification
Frontend Framework	React.js	19.x	Component-driven SPA with virtual DOM for blazing-fast UI updates
Build Tooling	Vite	8.x	Lightning-fast HMR development server with optimized production builds
Styling	Tailwind CSS	3.x	Utility-first responsive design with custom HSL color palettes
Routing (FE)	React Router DOM	6.x	Declarative route hierarchy with protected route guards

HTTP Client	Axios	1.7.x	Promise-based HTTP client with interceptors for auth headers
Charts	Chart.js + react-chartjs-2	4.x	Canvas-rendered bar/doughnut/line charts for admin analytics
Icons	Lucide React	0.395.x	Consistent, accessible SVG icon library
Backend Runtime	Node.js	18+	Server-side JavaScript enabling full-stack JS development
Web Framework	Express.js	4.x	Minimal, unopinionated RESTful API framework
Database	MongoDB + Mongoose	8.x	NoSQL document store with schema validation and ODM
Authentication	JWT + Bcrypt.js	—	Stateless token sessions + secure password hashing
Image Storage	Local uploads (Multer)	1.4.x	Disk-based image persistence with static file serving
Dev Fallback DB	mongodb-memory-server	11.x	In-memory MongoDB for development without external DB config
Dev Tools	Nodemon	3.x	Auto-restart server on file changes

2. Selection Rationales

- **Why MERN?** Full-stack JavaScript across database, backend, and frontend reduces context-switching and simplifies validation structures.

- **Why Vite over CRA?** Sub-second Hot Module Replacement (HMR) and highly optimized Rollup builds.
- **Why Local Multer over Cloudinary?** Zero external API credentials required for local staging, easily adaptable to S3/Cloudinary for production.